

VALUE STRATEGY CONSULTING

Technical Documentation

OCP Maintenance AI Platform - Documentacion Tecnica (AR)

Campo	Detalle
Document	DOC-OCP-2026-001
Date	March 2026
Version	1.0 - Final
Classification	Confidential - Internal Use
Scope	Full source code, Docker infrastructure, API
Methodology	Manual code review + static analysis
System	FastAPI + React 19 + PostgreSQL + Docker
Evaluated by	Value Strategy Consulting (VSC)

OCP Maintenance AI Platform

Built for reliability, designed to scale.

.....

-
1.
 2.
 3.
 4.
 5. FastAPI
 6. React + Vite
 7.
 8.
 9. API
 10. i18n
 11. SAP
 12. Docker
 13.
 14.
 15.
 16.
 17.
 18.

.....: 1.0.0
... ..: 2026
.....:
.....: OCP (.....)

.....

.....	
1	[.....](#1-.....)
2	[.....](#2-.....)
3	[.....](#3-.....)
4	[.....](#4-.....)
5	[..... FastAPI](#5-.....-fastapi)
6	[..... React + Vite](#6-.....-react--vite)
7	[.....](#7-.....)
8	[.....](#8-.....)
9	[..... API](#9-.....-api)
10	[..... i18n](#10-.....-i18n)
11	[..... SAP](#11-.....-sap)
12	[Docker](#12-docker-.....)
13	[.....](#13-.....)
14	[.....](#14-.....)
15	[.....](#15-.....)
16	[.....](#16-.....)
17	[.....](#17-.....)
18	[.....](#18-.....)

1.

.....

.... OCP
....., OCP
..... (JFC1).

.....

.....	
.....	 Weibull +
.... RCM	
.... SAP	 SAP PM
....	 (EN)· (ES)· (AR)
.....	
.....	

.....

.....		
1		
2	FMEA/FMECA RCM	
3		
4		

2.

.....

```
.....
.          ..... (.....)          .
.          React 19 SPA + Tailwind CSS          .
.....
.          ..... 8080          .
.          .          .
.....
.          Nginx (.... ....)          .
.          ..... 8080 (HTTP) / 8443 (HTTPS)          .
.          /api/* . FastAPI:8000 | /* . React:5173          .
.....
.          .          .
.          .          .
.....
.          FastAPI + Gunicorn          .          Vite + React 19          .
.          ..... 8000          .          ..... 5173          .
.....
.          .          .
.....
.          SQLite (.....)          .
.          PostgreSQL (.....)          .
.....
```

.....

.....		
..... (Vite)	5173	 React
..... (FastAPI)	8000	 REST
Nginx (HTTP)	8080	
Nginx (HTTPS)	8443	 SSL

.....

```
..... . Nginx:8080 . /api/* ..... . FastAPI:8000
. /* ..... . React:5173
```

3.

.....

.....		
Python	3.11	
FastAPI	0.133.1	
Uvicorn	0.41.0	 ASGI
Gunicorn	25.1.0	
SQLAlchemy	2.0.47	 (ORM)
Pydantic	2.12.5	
python-jose	3.3.0	 JWT
bcrypt	4.2.1	
Anthropic SDK	0.83.0	 (Claude)
openpyxl	3.1.5	 Excel
htpx	0.28.1	 HTTP

.....

.....		
React	19.1.0	
Vite	7.3.1	
Tailwind CSS	4.2.1	 CSS
React Router DOM	7.13.1	
Radix UI		
Recharts	3.7.0	
Lucide React	0.575.0	
XLSX	0.18.5	 Excel
file-saver	2.0.5	

.....

.....	
Docker	
Docker Compose	

4.

Nginx Alpine	
SQLite	
PostgreSQL	

```

ASSET-MANAGEMENT-SOFTWARE-master/
... api/                                # ..... (FastAPI)
.   ... main.py                         # .....
.   ... config.py                      # ..... (.....)
.   ... schemas.py                    # ..... Pydantic (..... 50 ..... )
.   ... seed.py                       # .....
.   ... database/
.       ... connection.py              # ..... SQLAlchemy .....
.       ... models.py                  # ..... ORM (..... 25 ..... )
.   ... routers/
.       ... auth.py                   # .....
.       ... hierarchy.py               # .....
.       ... criticality.py             # .....
.       ... fmea.py                    # FMEA/FMECA ..... RCM
.       ... tasks.py                   # .....
.       ... work_packages.py           # .....
.       ... sap.py                     # ..... SAP
.       ... analytics.py               # .....
.       ... work_requests.py           # .....
.       ... capture.py                 # .....
.       ... planner.py                  # .....
.       ... backlog.py                 # .....
.       ... scheduling.py              # .....
.       ... reliability.py              # .....
.       ... reporting.py                # .....
.       ... dashboard.py               # .....
.       ... rca.py                     # .....
.       ... admin.py                   # .....
.   ... services/                      # ..... (23 ..... )
.       ... auth_service.py            # ..... .JWT
.       ... hierarchy_service.py        # .....
.       ... hierarchy_builder_service.py # .....
.       ... criticality_service.py      # .....
.       ... fmea_service.py             # ..... FMEA/FMECA
.       ... task_service.py             # .....
.       ... work_package_service.py     # .....
.       ... sap_service.py              # ..... SAP
.       ... analytics_service.py        # .....
.       ... work_request_service.py     # .....
.       ... capture_service.py          # .....
.       ... planner_service.py          # .....
.       ... backlog_service.py          # .....
.       ... scheduling_service.py       # .....
.       ... reliability_service.py       # .....
.       ... reporting_service.py        # .....
.       ... rca_service.py              # .....
.       ... agent_service.py            # .....
.       ... capa_service.py             # .....
.       ... audit_service.py            # .....
.       ... validation_service.py       # .....
.   ... dependencies/
.       ... auth.py                    # .....
... frontend/                          # ..... (React + Vite)
.   ... package.json                   # .....
.   ... vite.config.js                  # ..... Vite
.   ... postcss.config.mjs              # ..... PostCSS
.   ... Dockerfile                      # .....
.   ... public/
.       ... OCP_LOGO.png                # ..... OCP
.   ... src/

```



```

.    ... main.jsx # .....
.    ... App.jsx # .....
.    ... api.js # .... API (.... .. 50 ....)
.    ... pages/ # ..... (23 ....)
.    .    ... Login.jsx # .....
.    .    ... Dashboard.jsx # .....
.    .    ... ExecutiveDashboard.jsx # .....
.    .    ... Hierarchy.jsx # .....
.    .    ... Criticality.jsx # .....
.    .    ... FMEA.jsx # .....
.    .    ... FMECA.jsx # .... FMECA
.    .    ... Strategy.jsx # .....
.    .    ... WorkRequests.jsx # .....
.    .    ... WorkPackages.jsx # .....
.    .    ... Backlog.jsx # .....
.    .    ... Scheduling.jsx # .....
.    .    ... Planning.jsx # .....
.    .    ... Planner.jsx # .....
.    .    ... FieldCapture.jsx # .....
.    .    ... Analytics.jsx # .....
.    .    ... Reports.jsx # .....
.    .    ... Reliability.jsx # .....
.    .    ... RCA.jsx # .....
.    .    ... DefectElimination.jsx # .....
.    .    ... SAPReview.jsx # ..... SAP
.    .    ... Admin.jsx # .....
.    .    ... Profile.jsx # .....
.    ... components/ # .....
.    .    ... Sidebar.jsx # .....
.    .    ... Header.jsx # .....
.    .    ... ErrorBoundary.jsx # .....
.    .    ... ConfirmDialog.jsx # .....
.    .    ... Toast.jsx # .....
.    .    ... Shared.jsx # .....
.    .    ... ProtectedRoute.jsx # .....
.    .    ... UpdateBanner.jsx # .....
.    .    ... ui/ # ..... Radix UI .....
.    .    ... accordion.jsx
.    .    ... alert-dialog.jsx
.    .    ... badge.jsx
.    .    ... button.jsx
.    .    ... card.jsx
.    .    ... checkbox.jsx
.    .    ... dialog.jsx
.    .    ... input.jsx
.    .    ... label.jsx
.    .    ... progress.jsx
.    .    ... scroll-area.jsx
.    .    ... select.jsx
.    .    ... separator.jsx
.    .    ... skeleton.jsx
.    .    ... table.jsx
.    .    ... tabs.jsx
.    .    ... textarea.jsx
.    .    ... tooltip.jsx
.    .    ... utils.js
.    ... contexts/ # ..... React
.    .    ... AuthContext.jsx # ..... (JWT)
.    .    ... LanguageContext.jsx # ..... (i18n)
.    ... i18n/ # .....

```

```

.      .    ... en.js                # .....
.      .    ... es.js               # .....
.      .    ... ar.js               # .....
.      ... data/
.      .    ... mockData.js         # .....
.      ... utils/
.      .    ... exportFile.js       # .....
.      ... styles/                  # ..... CSS .....
... sap_mock/                       # ..... SAP .....
.    ... data/
.      ... work_orders.json         # .....
.      ... functional_locations.json # .....
.      ... materials_bom.json       # .....
.      ... equipment_master.json    # .....
.      ... maintenance_plans.json   # .....
... Dockerfile                      # .....
... docker-compose.yml              # .....
... nginx.conf                      # .....
... requirements.txt                # ..... Python
... start.sh                        # .....

```

5. FastAPI

.....

... .. FastAPI create_app() api/main.py:

```
def create_app() -> FastAPI:
    app = FastAPI(
        title=settings.PROJECT_NAME,
        version="1.0.0",
        description="OCP Maintenance AI MVP · 4-module maintenance strategy platform",
        lifespan=lifespan,
    )
```

.....

1. FastAPI
2. CORS
3. API
4. 18 /api/v1
5.
6. /health

..... (18)

.....			
auth	/api/v1/auth	auth	
hierarchy	/api/v1/hierarchy	hierarchy	
criticality	/api/v1/criticality	criticality	
fmea	/api/v1/fmea	fmea	FMEA/FMECA RCM
tasks	/api/v1/tasks	tasks	
work_packages	/api/v1/work-packages	work-packages	
sap	/api/v1/sap	sap	 SAP
analytics	/api/v1/analytics	analytics	
work_requests	/api/v1/work-requests	work-requests	
capture	/api/v1/capture	capture	
planner	/api/v1/planner	planner	
backlog	/api/v1/backlog	backlog	
scheduling	/api/v1/scheduling	scheduling	
reliability	/api/v1/reliability	reliability	
reporting	/api/v1/reporting	reporting	
dashboard	/api/v1/dashboard	dashboard	

rca	/api/v1/rca	rca	
admin	/api/v1/admin	admin	

.....

..... SQLAlchemy ORM.

.....		
.....	auth_service.py	 JWT
.... FMEA	fmea_service.py	 RCM· FMECA
.....	analytics_service.py	 Weibull
.....	criticality_service.py	
.....	work_package_service.py	
.....	reporting_service.py	
.....	rca_service.py	 5W2H
.....	capture_service.py	
.....	agent_service.py	 Anthropic Claude
.....	hierarchy_service.py	
.....	hierarchy_builder_service.py	
.....	task_service.py	
.... SAP	sap_service.py	 SAP
.....	work_request_service.py	
.....	planner_service.py	
.....	backlog_service.py	
.....	scheduling_service.py	
.....	reliability_service.py	
.... CAPA	capa_service.py	
.....	audit_service.py	
.....	validation_service.py	

.....

..... api/config.py:

```
class Settings:
    DATABASE_URL: str = os.getenv("DATABASE_URL", "sqlite:///./ocp_maintenance.db")
    SAP MOCK_DIR: str = os.getenv("SAP MOCK_DIR", "sap_mock/data")
    ANTHROPIC_API_KEY: str = os.getenv("ANTHROPIC_API_KEY", "")
    API_V1_PREFIX: str = "/api/v1"
    PROJECT_NAME: str = "OCP Maintenance AI MVP"
    DEBUG: bool = os.getenv("DEBUG", "false").lower() == "true"
    CORS_ORIGINS: str = os.getenv("CORS_ORIGINS", "http://localhost:5173,...")
    JWT_SECRET_KEY: str = os.getenv("JWT_SECRET_KEY", "...")
    JWT_ALGORITHM: str = os.getenv("JWT_ALGORITHM", "HS256")
    JWT_ACCESS_TOKEN_EXPIRE_MINUTES: int = 240 # 4 .....
    JWT_REFRESH_TOKEN_EXPIRE_DAYS: int = 7 # 7 .....
```

6. React + Vite

..... (23

... .. frontend/src/main.jsx:

.....			
/login	Login	...	
/	Dashboard		
/executive-dashboard	ExecutiveDashboard	admin, manager	
/hierarchy	Hierarchy	admin, manager, engineer	
/criticality	Criticality	admin, manager, engineer	
/fmea	FMEA	admin, planner, engineer	
/fmeca	FMECA	admin, planner, engineer	 FMECA
/strategy	Strategy	admin, planner, engineer	
/work-requests	WorkRequests		
/work-packages	WorkPackages	admin, planner	
/backlog	Backlog	admin, planner	
/scheduling	Scheduling	admin, planner	
/planning	Planning	admin, planner	
/planner	Planner	admin, planner	
/field-capture	FieldCapture	admin, tecnico	
/analytics	Analytics	admin, manager, engineer	
/reports	Reports	admin, manager	
/reliability	Reliability	admin, engineer	
/rca	RCA	admin, engineer	
/defect-elimination	DefectElimination	admin, engineer	
/sap-review	SAPReview	admin, manager	 SAP
/admin	Admin	admin	
/profile	Profile		

.....

```

.....
· UpdateBanner ( ..... )
.....
·      · Header ( ..... )
·      · .....
· Sidebar · ..... | ..... | .....
· .....
· .....
· .....: .....
· #1B5E20 · ..... (Outlet)
· .....
· OCP      · ..... ErrorBoundary .....
· .....
· OCP      · .....
.....

```

.....

.....		
AuthContext	contexts/AuthContext.jsx	 JWT/.....
LanguageContext	contexts/LanguageContext.jsx	 t() RTL

.... API

... frontend/src/api.js 50:

- Bearer AuthContext
- 401: /login
- URL VITE_API_BASE_URL

```

// ..... API
export async function listPlants() {
  const res = await fetch(`${BASE}/api/v1/hierarchy/plants`, { headers: authHeaders() })
  return res.json();
}

```

.....

..... Radix UI .. Tailwind CSS:

- (Accessible)
- Tailwind
- 19 UI .. components/ui/

.....

*** *****	***** ***** *
*** *****	***** ***** *
***** *****	 OCP #1B5E20



7.

.....

..... 25 SQLite
..... PostgreSQL

.....

```
PLANT (....)
  ... AREA (.....)
    ... SYSTEM (....)
      ... EQUIPMENT (....)
        ... SUB_ASSEMBLY (.....)
          ... MAINTAINABLE_ITEM (.... . . . . .)
```

.....

.....		
UserModel		user_id, email, username, role, hashed_password
PlantModel		plant_id, name, name_fr, name_ar, location
HierarchyNodeModel		node_id, node_type, name, parent_node_id, plant_id, criticality
CriticalityAssessmentModel		assessment_id, node_id, criteria_scores, risk_class, overall_score
FunctionModel		function_id, node_id, function_type, description
FunctionalFailureModel		failure_id, function_id, failure_type, description
FailureModeModel		failure_mode_id, functional_failure_id, mechanism, cause, strategy_type
MaintenanceTaskModel		task_id, failure_mode_id, name, task_type, frequency_value
WorkPackageModel		work_package_id, name, node_id, frequency_value, status
WorkOrderModel		work_order_id, order_type, equipment_id, priority, status
HealthScoreModel		score_id, node_id, composite_score, health_class, trend
KPIMetricsModel		metrics_id, plant_id, mtbf_days, mtrr_hours, availability_pct
FailurePredictionModel		prediction_id, equipment_id, weibull_params, risk_score
SAPUploadPackageModel	 SAP	package_id, plant_code, maintenance_plan, status
CAPAItemModel	 CAPA	capa_id, capa_type, title, status, root_cause

.....

.....	
VarianceAlertModel	
ExpertCardModel	
AuditLogModel	
FieldCaptureModel	
WorkRequestModel	
PlannerRecommendationModel	
BacklogItemModel	
OptimizedBacklogModel	
WorkforceModel	
ShutdownCalendarModel	
InventoryItemModel	
WeeklyProgramModel	
ShutdownEventModel	
MoCRequestModel	
SparePartAnalysisModel	
RBIAssessmentModel	
ReportModel	
NotificationModel	
RCAAnalysisModel	
PlanningKPISnapshotModel	
DEKPISnapshotModel	
UserFeedbackModel	
FMECAWorksheetModel	 FMECA

8.

.... JWT

```
1. .... . POST /api/v1/auth/login (username + password)
2. .... . { access_token (4 ....), refresh_token (7 ....) }
3. .... . Authorization: Bearer <access_token>
4. .... . 401 . POST /api/v1/auth/refresh
5. .... . .... /login
```

.....

....		
admin		
manager		
planner		FMEA.
tecnico		
engineer		 FMEA.

.....

... ..		
admin	admin123	
manager	manager123	
planner	planner123	
tecnico	tecnico123	...

.....

-: bcrypt
-: (hashed_password)
-: bcrypt.checkpw()

9. API

..... (`/api/v1/auth`)

.....			
POST	/auth/login	 JWT	..
POST	/auth/refresh		..
POST	/auth/register		admin
GET	/auth/me		...
PUT	/auth/me		...
PUT	/auth/change-password		...
GET	/auth/users		admin
PUT	/auth/users/{user_id}/role		admin
PUT	/auth/users/{user_id}/deactivate		admin
PUT	/auth/users/{user_id}/activate		admin

..... (`/api/v1/hierarchy`)

.....			
GET	/hierarchy/plants		..
POST	/hierarchy/plants		...
GET	/hierarchy/nodes	 (..)	..
POST	/hierarchy/nodes		...
GET	/hierarchy/nodes/{node_id}		..
GET	/hierarchy/nodes/{node_id}/tree		..
POST	/hierarchy/build-from-vendor		...
GET	/hierarchy/stats		..

FMEA (`/api/v1/fmea`)

.....			
POST	/fmea/failure-modes		...
GET	/fmea/failure-modes/{fm_id}		..
GET	/fmea/failure-modes		..
GET	/fmea/validate-fm	 /...	..

GET	/fmea/fm-combinations		..
POST	/fmea/rcm-decide	 RCM	..
GET	/fmea/functions		..
POST	/fmea/functions		...
GET	/fmea/functional-failures		..
POST	/fmea/functional-failures		...
GET	/fmea/fmea/worksheets	 FMECA	..
POST	/fmea/fmea/worksheets	 FMECA	...
GET	/fmea/fmea/worksheets/{id}	 FMECA	..
POST	/fmea/fmea/rpn		..
PUT	/fmea/fmea/worksheets/{id}/run-decisions	 FMECA	...
GET	/fmea/fmea/worksheets/{id}/summary	 FMECA	..

..... (`/api/v1/analytics`)

.....			
POST	/analytics/health-score		...
POST	/analytics/kpis		...
POST	/analytics/weibull-fit	 Weibull	..
POST	/analytics/weibull-predict	 Weibull	...
POST	/analytics/variance-detect		..
GET	/analytics/variance-alerts		..
GET	/analytics/asset-health		..

..... (`/api/v1/reporting`)

.....			
POST	/reporting/reports/weekly		...
POST	/reporting/reports/monthly		...
POST	/reporting/reports/quarterly		...
GET	/reporting/reports		..
GET	/reporting/reports/{report_id}		..
POST	/reporting/de-kpis/calculate		...
POST	/reporting/de-kpis/program-health	 DE	...
POST	/reporting/notifications/generate		...

GET	/reporting/notifications		..
PUT	/reporting/notifications/{id}/ack		...
POST	/reporting/import/validate		...
POST	/reporting/export		...
POST	/reporting/cross-module/analyze		...

10. i18n

.....

.....			
.....	en	 (LTR)	frontend/src/i18n/en.js
.....	es	 (LTR)	frontend/src/i18n/es.js
.....	ar	 (RTL)	frontend/src/i18n/ar.js

..... .. React Context

```
// ..... LanguageContext
import { useLanguage } from '../contexts/LanguageContext';

function MyComponent() {
  const { t, language, setLanguage } = useLanguage();
  return <h1>{t('dashboard.title')}</h1>;
}
```

.....

.....;

.....		
.....	common	 (....)
.....	auth	
.....	nav	
.....	hierarchy	
.....	criticality	
FMEA	fmea	
.....	tasks	
... ..	workPackages	
.....	analytics	
.....	reports	
.....	planner	
.....	strategy	
.....	admin	

... RTL

-;
 - dir="rtl" <html>
 - (Sidebar)
 -
-

11. SAP

.....

.... SAP sap_mock/data/:

.....	 SAP	
work_orders.json	IW31/IW32	
functional_locations.json	IL01	
materials_bom.json	CS01	
equipment_master.json	IE01	
maintenance_plans.json	IP10	

.... SAP

1. (Work Package)
.
2.
.
3. SAP
.
4. SAPReview
.
5.
.
6. SAP PM (.....)

.... SAP

... ..	... SAP	
work_order_id	AUFNR	
equipment_tag	EQUNR	
task_type	AUART	
priority	PRIOK	
description	KTEXT	
planned_hours	ARBEI	

12. Docker

.....

.....			
ocp-backend	FastAPI + Gunicorn	8000	
ocp-frontend	React + Nginx	5173	
ocp-nginx	Nginx Alpine	8080 / 8443	

.....

.....

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
CMD ["gunicorn", "api.main:app", "-k", "uvicorn.workers.UvicornWorker", "-b", "0.0.0.0:8000"]
```

.....

```
# .....
FROM node:20-alpine AS build
WORKDIR /app
COPY package*.json .
RUN npm ci
COPY . .
RUN npm run build

# .....
FROM nginx:alpine
COPY --from=build /app/dist /usr/share/nginx/html
```

..... Docker

.....	
docker-compose up -d --build	
docker-compose logs -f ocp-backend	
docker-compose stop	
docker-compose down	
docker-compose up -d --build ocp-backend	

.....

```
networks:
  ocp-network:
    name: ocp-maintenance-net
    driver: bridge

volumes:
  ocp_db_data:      # .....
  ocp_sap_data:     # ..... SAP .....
```

13.

1:

.....		
.....	Hierarchy.jsx	 (.....)
.....	Criticality.j	 (..... ×

-:
-/OEM
 -: A (···)· B (···)· C (···)
 -
 - SAP

2: FMEA/FMECA RCM

.....		
FMEA	FMEA.jsx	 RCM
FMECA	FMECA.jsx	 FMECA .. RPN (..... × ×
.....	Strategy.jsx	

.....:

.....		
CBM		
TBM		
FTM		 (.....)
RTF		 (.....)

3:

.....		
.....	WorkRequests.jsx	
... ..	WorkPackages.jsx	
.....	Backlog.jsx	
.....	Scheduling.jsx	
.....	Planning.jsx	
.....	Planner.jsx	
.....	FieldCapture.jsx	 (··/··/···)

..... 4:

.....		
.....	Analytics.jsx	
.....	ExecutiveDashboard.jsx	
.....	Reports.jsx	/...../.....
.....	Reliability.jsx	 Weibull· RBI·
.....	RCA.jsx	 5W2H
.....	DefectElimination.jsx	 CAPA
..... SAP	SAPReview.jsx	 SAP

14.

1.

.....
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..... (..... ×)
.
.....
.
.....
.
.....
.
..... (A/B/C)

2. FMEA

.....
.
.....
.
..... (..... +)
.
..... RCM
.
..... (CBM/TBM/FTM/RTF)
.
.....
.
.....
.
..... SAP

3.

..... (.. / .. / ..)
.
..... (.....)
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.....
.
..... AI
.
..... / / /
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4.

..... (...../...../.....)
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.....
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..... (Excel / PDF)

15.

.....

.....	
..... JWT	 HS256. 4 7
.....	bcrypt
..... CORS	
..... API	... X-API-Key (POST/PUT/DELETE)
.....	30 .../..... 2 .../.....
.....	X-Content-Type-Options, X-Frame-Options, X-XSS-Protection ... Nginx
..... HTTPS	... SSL/TLS ... Nginx ... 8443
.....	5
.....	 (non-root)
..... .. SQL	 SQLAlchemy ORM (parameterized)
..... CSRF	 SameSite

16.

.....

.....	
Python	3.11
Node.js	20
npm	9
Git	

.....

```
# 1. ....
python -m venv venv

# 2. ....
# .. Windows:
venv\Scripts\activate
# .. Linux/macOS:
source venv/bin/activate

# 3. ....
pip install -r requirements.txt

# 4. .... .env (... ..)
cp .env.example .env
# ..

# 5. ....
uvicorn api.main:app --reload --port 8000
```

.....

```
# 1. ....
cd frontend

# 2. ....
npm install

# 3. ....
npm run dev
```

.....

.....		
.....	http://localhost:5173	 React
.....	http://localhost:8000	 API
..... API	http://localhost:8000/docs	 Swagger UI
...	http://localhost:8000/health	

.....

```
# ... curl
curl -X POST http://localhost:8000/api/v1/admin/seed

# .. ... Python
python -c "from api.seed import seed_database; seed_database()"
```

17.

.....;

.....			
1	 JWT		 JWT_SECRET_KEY ... (256)
2	PostgreSQL		 SQLite ... PostgreSQL ... DATABASE_URL
3	HTTPS		 SSL/TLS ... Nginx ... 8443
4	 SSL		 SSL /certs/
5	 API		 API_KEY
6	 CORS		 CORS_ORIGINS
7	 Anthropic		 ANTHROPIC_API_KEY
8			 DEBUG=false
9			 Nginx
10	 Docker		 (ocp_db_data)
11			 /health
12			 docker-compose up -d --build

18.

.....

.....		
DATABASE_URL	sqlite:///./ocp_maintenance.db	 (SQLite Pos
SAP MOCK DIR	sap_mock/data	 SAP
ANTHROPIC_API_KEY	"" (....)	 API Anthropic Claude
GOOGLE_AI_API_KEY	"" (....)	 API Google AI
API_KEY	"" (....)	 API (.....)
DEBUG	false	 (... .. false
CORS_ORIGINS	http://localhost:5173,http://localhost:8501,	 CORS (..... ..)
LOG_LEVEL	INFO	 (DEBUG, INFO, WARNING, ERROR)
JWT_SECRET_KEY	ocp-maintenance-dev-secret-key-change-in-pro	 JWT (... ..
JWT_ALGORITHM	HS256	 JWT
JWT_ACCESS_TOKEN_EXPIRE	240	 (4
JWT_REFRESH_TOKEN_EXPIRE	7	
BACKEND_PORT	8000	
FRONTEND_PORT	5173	
NGINX_PORT	8080	 HTTP .. Nginx
NGINX_SSL_PORT	8443	 HTTPS .. Nginx
PYTHONPATH	/app	 Python
VITE_API_BASE_URL	"" (....)	 URL API

.... .. `env`

```
# .....  
DATABASE_URL=postgresql://ocp_user:ocp_password@db:5432/ocp_maintenance  
  
# .....  
JWT_SECRET_KEY=your-very-secure-random-secret-key-256-bits-minimum  
JWT_ALGORITHM=HS256  
JWT_ACCESS_TOKEN_EXPIRE_MINUTES=240  
JWT_REFRESH_TOKEN_EXPIRE_DAYS=7  
  
# .....  
ANTHROPIC_API_KEY=sk-ant-xxxxxxxxxxxxx  
  
# .....  
API_KEY=your-api-key-for-mutations  
CORS_ORIGINS=https://your-domain.com  
DEBUG=false  
LOG_LEVEL=INFO  
  
# .....  
BACKEND_PORT=8000  
FRONTEND_PORT=5173  
NGINX_PORT=8080  
NGINX_SSL_PORT=8443
```

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.....	
..... API	18
.... API	 100
.....	 25
.....	23
..... React	23
.....	 20
..... Pydantic	 50
.... API	 50
.....	3 (.....
..... Docker	3
..... Python	27
..... npm	 30

.... OCP -- OCP (.....)
..... 2026

OCP Maintenance AI Platform

Documentacion Tecnica Completa

Evaluated by Value Strategy Consulting

Fin del documento

Este documento contiene informacion tecnica sobre la plataforma OCP Maintenance AI. Su distribucion esta restringida al equipo de desarrollo y direccion.